

STANDARD DATABASE ENGINE FOR BANKING AND OFFICE TEST

AiresDB v0.1.0 vs Firebird 5.0.4

SDEBO 1.0 publication run - SDEBO-S750 - seed 1999 - Windows embedded profile - 9-11
September 2026



Decision

Engine	Medium Office	Small Bank Technical	Critical finding
AiresDB	PASS	PASS	R01 capped at BC: offline checkpoint + file copy
Firebird	PASS	PASS	Q07 full scan is below target B

Both engines pass the applicable Medium Office and Small Bank Technical correctness gates in this embedded profile. AiresDB's final score is held back mainly by the absence of a native backup format. Firebird's main measured weakness is full-scan throughput. Scores apply only to this hardware, configuration, dataset, and harness.

Method and fairness

The public SDEBO 1.0 procedure was followed with the full 750,000-row dataset. The generator used seed 1999 and produced the same pipe-delimited source files for both engines. Each engine received a fresh database for each of five independent performance runs. Point operations used 100 warm-up and 1,000 measured iterations; scans, aggregates, and joins used 2 warm-up and 7 measured iterations; writes used 20 warm-up and 200 measured operations. Final performance values are medians across the five independent runs.

Durability was enabled for both products: AiresDB committed through its fsync-backed WAL and Firebird used forced writes in synchronous mode. Money was stored as signed 64-bit integer cents. Both products ran embedded on the same OS and SSD. No performance run overlapped another performance run.

Environment

Field	Recorded value
OS	Windows-10-10.0.19045-SP0
CPU	Intel(R) Core(TM) i5-2400S CPU @ 2.50GHz
CPU topology	4 physical / 4 logical
RAM	7.97 GiB
Storage	SSD; 31.29 GiB free after test
Dataset	SDEBO-S750; 750,000 rows; seed 1999
Source digest	abb9654f1a69e5cd47b0ce546116b377ca59d3ec0b83077a7a48d1d2a8669799
AiresDB	0.1.0; Julia 1.12.7; embedded; WAL fsync per commit
Firebird	5.0.4.1812; firebird-driver 2.0.3; embedded; forced writes sync

Comparison limits

The host has four hardware threads, below the comparison recommendation of four cores/eight threads, but above the published minimum. VM hard power-off was unavailable; each engine instead completed five externally terminated process cycles at transaction boundaries. S01 server/network gates are outside the embedded profile. These deviations are disclosed rather than silently substituted.

Performance results

PFR compares the observed median with the published B target. Throughput uses observed/target; latency uses target/observed. A requires PFR ≥ 1.50 , AB ≥ 1.20 , B ≥ 1.00 , BC ≥ 0.75 , C ≥ 0.50 , D ≥ 0.25 .

Test	Target B	AiresDB median	Grade	Firebird median	Grade	Observed comparison
Q01 Bulk load	1,000 rows/s	23,103 rows/s	A	1,809 rows/s	A	AiresDB 12.77x Firebird
Q02 PK lookup p95	5.000 ms	2.522 ms	A	0.406 ms	A	Firebird 6.21x lower latency
Q03 Indexed filter p95	15.000 ms	2.538 ms	A	0.311 ms	A	Firebird 8.17x lower latency
Q04 Range p95	25.000 ms	2.470 ms	A	3.864 ms	A	AiresDB 1.56x lower latency
Q05 Aggregate median	1,500.000 ms	1,094.064 ms	AB	995.906 ms	A	Firebird 1.10x lower latency
Q06 Join median	2,000.000 ms	27.884 ms	A	5.489 ms	A	Firebird 5.08x lower latency
Q07 Full scan	50,000 rows/s	159,819 rows/s	A	41,400 rows/s	BC	AiresDB 3.86x Firebird
T01 Insert commit p95	10.000 ms	3.653 ms	A	1.908 ms	A	Firebird 1.91x lower latency
T02 Update/delete p95	15.000 ms	3.491 ms	A	1.626 ms	A	Firebird 2.15x lower latency

AiresDB leads bulk loading by about 12.8x and full scanning by about 3.9x. Firebird has lower point lookup, indexed filter, aggregate, join, and committed-write latency on this host. AiresDB still exceeds every published B performance target; its aggregate receives AB because the median is 1.37 times the target rather than the 1.50 required for A.

Transactions, recovery, and security

Test	Weight	AiresDB	AiresDB evidence	Firebird	Firebird evidence
Q08	3	A	16/16 controls/families	A	16/16 controls/families
T03	4	A	1000 transfers plus 5 crash boundaries	A	1000 transfers plus 5 crash boundaries
T04	4	A	200 rollback cycles preserved state	A	200 rollback cycles preserved state
T05	4	A	4 workers; 0 lost updates	A	4 workers; 0 lost updates
R01	3	BC	3 usable offline copies; non-native cap	A	3 native gbak backups usable
R02	4	B	3 manually orchestrated restores match	A	3 automated native restores match
R03	4	A	5 process-kill cycles valid; VM unavailable	A	5 process-kill cycles valid; VM unavailable
R04	4	A	reopen, recovery, and index checks passed	A	reopen, recovery, and index checks passed
R05	4	A	3 corruptions detected or safely recovered; no silent mismatch	A	3 corruptions detected or safely recovered; no silent mismatch
S01	3	C	5/8 controls/families	AB	7/8 controls/families
O01	2	B	memory growth 13.90%; 900.9s; 0 errors	A	memory growth 0.00%; 900.6s; 0 errors

All tested atomicity, rollback, concurrency, restore-digest, crash-reopen, journal/index, and corruption-safety checks completed without a silent mismatch. Firebird used three native gbak backups and restores. AiresDB v0.1.0 has no native backup command, so the test used a quiescent checkpoint followed by a documented file copy; the public rule caps that result at BC.

Corruption tests deliberately modified a header, a data/page region, and a tail/log region on disposable copies. A pass means the engine rejected the copy or reopened to the exact pre-recorded logical digest. It does not claim that every possible on-disk corruption pattern was exhaustively tested.

Sustained workload

Metric	AïresDB	Firebird
Duration	900.9 s	900.6 s
Operations	20,298	19,797
Errors	0	0
RSS start	796.6 MiB	53.3 MiB
RSS end	907.3 MiB	46.7 MiB
RSS peak	986.8 MiB	69.4 MiB
Average process CPU	87.4% of one logical core	91.7% of one logical core
Memory growth	13.90%	-12.43%
Latency p95	709.418 ms	26.545 ms
Last/first-quarter latency	0.920x	1.014x
Disk growth	1.4 MiB	0.0 MiB

The 15-minute mixed workload combined indexed lookups, committed updates, 100-row range queries, and grouped aggregates. O01 grading follows end-to-start RSS growth and requires zero operation errors plus a successful restart. Peak RSS is reported separately because temporary working memory can be reclaimed before the end of the run.

Complete weighted score

Code	Wt	AiresDB grade	Pts	Weighted	Firebird grade	Pts	Weighted
Q01	3	A	4.0	12.0	A	4.0	12.0
Q02	4	A	4.0	16.0	A	4.0	16.0
Q03	3	A	4.0	12.0	A	4.0	12.0
Q04	3	A	4.0	12.0	A	4.0	12.0
Q05	3	AB	3.5	10.5	A	4.0	12.0
Q06	4	A	4.0	16.0	A	4.0	16.0
Q07	2	A	4.0	8.0	BC	2.5	5.0
Q08	3	A	4.0	12.0	A	4.0	12.0
T01	3	A	4.0	12.0	A	4.0	12.0
T02	4	A	4.0	16.0	A	4.0	16.0
T03	4	A	4.0	16.0	A	4.0	16.0
T04	4	A	4.0	16.0	A	4.0	16.0
T05	4	A	4.0	16.0	A	4.0	16.0
R01	3	BC	2.5	7.5	A	4.0	12.0
R02	4	B	3.0	12.0	A	4.0	16.0
R03	4	A	4.0	16.0	A	4.0	16.0
R04	4	A	4.0	16.0	A	4.0	16.0
R05	4	A	4.0	16.0	A	4.0	16.0
S01	3	C	2.0	6.0	AB	3.5	10.5
O01	2	B	3.0	6.0	A	4.0	8.0
FINAL	68	Excellent	3.74	-	Excellent	3.93	-

Gate outcome

Engine	Medium Office	Small Bank Technical	Critical finding
AiresDB	PASS	PASS	R01 capped at BC: offline checkpoint + file copy
Firebird	PASS	PASS	Q07 full scan is below target B

For embedded operation, S01 is scored for disclosure but its server/network gate is not applied. Small Bank Technical PASS requires final score above 3.00, grades of at least B on T03, T04, R02, R03, and R05, plus no silent corruption, half transaction, or backup mismatch.

Evidence and reproducibility

The publication directory includes result.json, result.csv, environment.json, raw performance JSON for all ten main runs, resilience evidence for both engines, the deterministic dataset manifest, harness source hashes, and SHA256SUMS. Database files are retained in the run directories for local audit but are excluded from the compact publication checksum list when redundant.

Source references

Benchmark rules and scoring: SDEBO Test Specification v1.0 publication document supplied with this repository. Comparator distribution: Firebird 5.0.4.1812 Windows x64 archive, official release. Driver: firebird-driver 2.0.3. Firebird embedded access, forced writes, gbak, and gfix behavior follow the Firebird 5 Quick Start Guide and bundled command-line help.

Official links

Firebird 5.0 release page: <https://firebirdsql.org/en/firebird-5-0/>

Firebird 5 Quick Start Guide: <https://www.firebirdsql.org/file/documentation/html/en/firebirddocs/qsg5/firebird-5-quickstartguide.html>

Python driver documentation: <https://github.com/FirebirdSQL/python3-driver/blob/master/docs/getting-started.txt>

Interpretation

This report is a single-host technical comparison, not a universal product ranking. AiresDB's strongest measured advantages are bulk ingestion and sequential scanning; Firebird's mature page engine delivers lower latency in most point and report operations and provides native backup tooling. The result supports continued AiresDB work on aggregate latency, memory footprint, native online backup, and server security while preserving its current high-throughput data path.

Additional engineering observation

An unscored first AiresDB T05 attempt found a Session lifecycle defect: closing one independently constructed worker Session while other workers were still committing closed the shared page manager and produced Commit Outcome Unknown. The scored rerun retained every worker until all measured operations completed, then recorded 1,000 commits with no lost update. The lifecycle behavior remains a stability issue to fix and is preserved in raw/harness-notes. A separate first O01 attempt was invalidated solely for an incorrect harness update_key! call and was rerun for the full duration.